

Karthikheyaa Kurra

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EDUCATION

University of Delaware <i>M.S. Data Science, GPA 3.80/4.00</i> • Coursework: Probability Theory, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering	Newark, DE Aug 2024 – May 2026
Sathyabama Institute of Science & Technology <i>B.E. Computer Science & Engineering</i>	Chennai, India Oct 2020 – May 2024

EXPERIENCE

Research Associate, Scientific Machine Learning <i>Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)</i>	Newark, DE Jun 2026 – Present
• Writing a paper on hybrid PDE solvers that combine classical methods and neural networks to solve nonlinear elliptic PDEs non-iteratively via learned surrogates and hierarchical structure. • Built high-precision ground truth with Chebyshev spectral discretizations and Newton iteration, plus dataset validation and reproducible PyTorch training/evaluation pipelines. • Ran ablation studies on network depth, normalization, and physics-informed residual losses with adaptive loss balancing; profiled inference complexity and wall-clock time. • Maintained the research codebase with unit/smoke tests and weekly progress reports.	
Research Software Engineer, LLM Serving & Benchmarking <i>Department of Computer & Information Sciences, UD (Advisor: Dr. Sunita Chandrasekaran)</i>	Newark, DE Jun 2026 – Present
• Part of a team building an on-premise LLM assistant for UD students, faculty, and staff. • Benchmarking LLM inference/serving (throughput, latency, model selection) on NVIDIA DGX hardware.	
Research Assistant, Scientific Computing <i>Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)</i>	Newark, DE Jun 2025 – Jun 2026
• Built a hierarchical Poincaré–Steklov (HPS) solver for linear elliptic PDEs: Chebyshev leaf grids, Dirichlet-to-Neumann maps, and Schur-complement merges up the tree. • Verified on manufactured solutions (Laplace, Poisson, Helmholtz) and wrote notes on leaf operators, merges, and hierarchy scaling.	
Teaching Assistant, Machine Learning <i>Sathyabama Institute of Science & Technology</i>	Chennai, India Jan 2024 – Apr 2024
• Graded assignments, midterms, and homework; prepared lecture slides and notes for the undergraduate ML course. • Led review sessions on mathematics for machine learning (matrix algebra and multivariable calculus).	
Research Intern, Computer Vision <i>Indian Institute of Technology, Hyderabad</i>	Hyderabad, India Mar 2023 – Jun 2023
• Applied supervised contrastive learning (Khosla et al., 2020) to facial expression recognition in PyTorch, comparing SupCon against a cross-entropy baseline. • Analyzed embeddings with t-SNE and delivered a reproducible codebase to the lab.	
Summer Intern <i>The University of Texas at Dallas</i>	Dallas, TX Jun 2022 – Jul 2022
• Summer program in machine learning and deep reinforcement learning: lectures, labs, and model building from first principles.	

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL, Bash, LaTeX
Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)
Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

READING GROUPS

Scientific Machine Learning Reading Group, University of Delaware (Advisor: Dr. Ke Chen), 2025–Present.
Weekly student presentations; topics include Computational Optimal Transport (Peyré & Cuturi), Equivariant Neural Networks, and One-Shot Learning.